

CCT Campaign Plan — Fix $\hat{A}_i$, Find the Squid vs Jackal

Prepared by COMPASS (for Charles, SCOUT, Alex)
Date: 2026-08-21 (evening refresh)
Status: Act II empirical core complete — P1 grid (min10/min20) · P3 · PPM triptych · P2b Alex board (K/N tail bins) shipped. +DFT PPM CCT whisper on poolq_loo; **flat-then-down on T_j ** at +DFT (P2b). Next: Act III — HAND framing + paper outline (Alex, post-next week: 50% writing / 50% code-data). P2 heatmap parked. SCOUT review: Complete (2026-08-21) — 20260821_SCOUT_CCT_campaign_review.md
Canonical companions (do not re-derive from memory):

Doc	Role
20260820_COMPASS_Charles_CCT_porch_reading.md	Narrative + science (SCOUT approved 2026-08-21)
20260821_SCOUT_porch_reading_review.md	Spot-checks, three clarifications
../HEROs_and_PASSes/PD20_22_campaign_big_picture.md	Dissertation arc (Army → MBB ladder)
../../BINDING_Selection_is_its_own_step.md	Score $\neq$ select $\neq$ environment

0 — One sentence (what Alex just got excited about)

Hold individual talent $\hat{A}_i$ fixed. Then ask: at the same fish size, does draft probability fall when the player moves from a Squid pond (stands out on a pretty-good team) to a Jackal pond (same talent, crowded among contenders)?
That is the Central Contention and Theme (CCT). Alex’s reaction — fix A_i — is exactly the microscope this campaign builds. Marginal hero curves average over ability; they cannot answer this question. Conditional plots can.

1 — Campaign arc (three acts)

flowchart LR; A[Act I – BDP population geometry] --> B[Act II – Conditional CCT empirics]; B --> C[Act III – HAND deck + Alex defense]; A -. -> IDONEI A; B --> D[Act IIb – Sensitivity optional]; D --> C;

Act	Question	Status	Owner
I — BDP	What do the pools look like? ($\hat{A}$, T_j , roster size, +DFT)	Closed (2026-08-20)	Charles + scripts
II — Conditional	At fixed $\hat{A}$, does draft rate fall as pond thickens?	Empirics shipped — poolq_loo + T_j + rank; see §3	Charles (HAND) · COMPASS (wording)
III — HAND + Alex	Can you show and defend CCT in one deck?	Starting — Alex on track; story framing after next week	Charles (slides) · COMPASS (wording)

Win condition (revised 2026-08-21 evening): A defensible HAND story — three magnifiers, not one bar:
1. poolq_loo P1: full panel NO; +DFT PPM whisper (~2 pp at [1.5,2]; stronger at [2,3] triptych — bin 16 cliff).
2. P3: within-team rank on contenders (Q4 top 9.5% vs mid 3.0%).
3. **P2b T_j (Alex board):** +DFT flat-then-down at fixed $\hat{A}$ with tail binning (primary z [2,3]: plateau ~34% → tail ~8%).
Alex (2026-08-21): on track; more done since summer start than prior two years; after next week → frame story, outline/write paper (placeholders OK), then 50% writing / 50% code-data cleanup.
Not the win condition: Claiming CCT confirmed from any single plot; OBPM canonical; ignoring thin tail cells on P2b.

2 — Scientific guardrails (binding)

- Environment ($L_{net} = B - D$) $\neq$ advancement. CCT is a draft / selection story, not “bad teammates hurt my box score.”
 - Score ($S_i = A_i - \lambda L_C$) $\neq$ select (top-K, Gibbs). Hero is an outcome; do not merge layers in talk or captions.
 - Congestion axis: poolq_loo first (“teammates around me,” excluding self). ** T_j ** second only — team mean includes the player.
 - Locked panel for claims: mg10 min20 11_21 (same estimand as POST-QC hero). QC always on: drop_dash_placeholder_names=True (dash “_” rows). FP / min0 = QC story only.
 - Perf metric: PPM canonical; BPM / OBPM = robustness appendix (Track C: no elite dip rescue).
 - +DFT overlay: magnifier on draft-ecosystem subsample — not “drafted players only” on player histograms.
- Army cross-read (do not garble): Army screams (robust macro tail drop). Captain promotion ~35–40% — assembly-line baseline, not tiny K/N. NCAA draft ~2–2.5%. Army loud $\neq$ Army slots scarcer than NBA picks.

3 — Plot menu (priority order)

Build in order unless Charles reorders. Each plot ships PNG + JSON sidecar (cell counts mandatory).

Priority 1 — Matched $\hat{A} \times$ pond (THE Alex plot) ★

Spec

- Panel: mg10 min20 11_21
- Fix narrow $\hat{A}_i$ band (default: $z \in [1.5, 2.0]$; sensitivity: top decile)
- X-axis: **poolq_loo** ventiles (preferred) or T_j ventiles
- Y-axis: draft rate (with binomial CI if n allows)
- **poolq winsor**: 0.01–0.99 (match locked POST-QC hero — BDP $\hat{A}/T$ plots omit winsor; this plot does not)
- Squid proxy: mid pond / mid T_j · Jackal proxy: top pond / top T_j — **same $\hat{A}$ band**

CCT signature: Jackal bar **below** Squid bar at same individual talent.

Outputs

- `basic_data_plots/CCT_draft_rate_ai_band_poolq_loo.png`
- `basic_data_plots/CCT_draft_rate_ai_band_poolq_loo.json`

Priority 1 scorecard (2026-08-21) — all variants shipped

Spec locked: mg10 min20 2011–2021 · PPM/OBPM $z \in [1.5, 2.0]$ · poolq_loo ventiles (within band) · winsor 0.01–0.99 · Squid = bins 6–8 · Jackal = bins 14–16.

Variant	Squid	Jackal	CCT (Squid > Jackal)?	Files
PPM · full panel	7.3% (n=382)	10.8% (n=381)	NO	<code>CCT_draft_rate_ai_band_poolq_loo.*</code>
PPM · +DFT	22.9% (n=140)	20.9% (n=139)	YES (~2 pp; CIs overlap)	<code>..._poolq_loo_dft.*</code>
OBPM · full	16.3% (n=306)	37.6% (n=306)	NO (strong opposite)	<code>..._poolq_loo_obpm.*</code>
OBPM · +DFT	29.0% (n=114)	39.8% (n=113)	NO	<code>..._poolq_loo_obpm_dft.*</code>
PPM · T_j axis (8 bins)	—	—	Draft rate rises with T_j at fixed $\hat{A}$	<code>CCT_draft_rate_fixedAi_Tj_*_Tj.*</code>

Band (PPM full): 2,035 PS / 178 drafted (~8.8% in band; **166/178 drafts** on +DFT teams). All 16 ventiles $n \approx 127$ –128 (full) or ~46 (+DFT).

[COMPASS] synthesis

1. **Full league**: thicker poolq_loo → **higher** draft rate at fixed $\hat{A}$ — opposite naive CCT on this axis. Confound: elite programs raise both pond quality and draft visibility.
2. **+DFT PPM**: sign **flips** toward CCT — BDP “draft-ecosystem zoom” intuition validated, but effect is **small** and **noisy** (bin 14 spikes; bins 15–16 fall; pooled proxies hide ventile mess).
3. **OBPM**: wrong magnifier — Jackal dominates even more; **appendix only** (Track C).
4. **** T_j secondary****: monotonic rise with team mean — “better team helps” at fixed z .
5. **Priority 3** (below): different question — **within-team rank** on Q4 contenders (top 9.5% vs mid 3.0%).

Honest Alex line (post-sweep): “Full panel at fixed talent: crowded ponds draft more. Restrict to programs that actually produce picks, PPM sign flips — mid-pond edges top-pond by ~2 points but uncertainty overlaps. OBPM goes the other way. On contenders, you must be near the top of your roster.”

HAND lead deck (2026-08-21):

Slide	File	Role
Triptych (poolq_loo)	<code>CCT_draft_rate_ai_band_poolq_loo_min10_ppm_triptych.png</code> (+ <code>_b8</code>)	Band sign flip [1,2]→[2,3]; +DFT bin-16 cliff
Alex board (T_j)	<code>CCT_draft_rate_fixedAi_Tj_knbins_dft.png</code>	Flat-then-down at fixed $\hat{A}$ (+DFT)
P3 Q4	<code>CCT_draft_rate_roster_pct_by_tj_quartile.png</code>	Stand out on your team

Side-by-side **PPM full vs +DFT** (locked [1.5,2] min20) still valid for “league confound” beat.

Priority 2 — $\hat{A} \times$ poolq_loo heatmap — PARKED

Draft rate in every cell ($\hat{A}$ ventile $\times$ poolq_loo ventile). Useful for smearing diagnosis; **not** Alex’s re-raised ask (PD25 = P2b). Build if time after Act III framing.

Priority 2b — Alex board plot: fixed $\hat{A} \times T_j$ (K/N bins) — SHIPPED ★

Source: PD25 (2026-08-21) — `../../../../../transcripts/PD25_notes.md` · board `../../../../../transcripts/PD25_board.jpeg` · plain-English: `../DISPOSABLE_PD25_Alex_board_for_dummies.md`

Question (Alex): At fixed individual talent, does draft rate stay **flat** in team talent, then **turn down** once congestion kicks in near K/N?
Implementation (2026-08-21): `pass_a_congestion_conditional.py --plot fixed_ai_tj_knbins` · **piecewise tail** (4 coarse + 20 fine bins on T_j within band) · T_j includes self (not poolq_loo).

P2b scorecard

Variant	Plateau (low/mid bins)	Tail (last 3 bins)	Downturn?	Files
PPM z [2,3] · +DFT · min10 (primary)	~34%	~8%	YES	CCT_draft_rate_fixedAi_Tj_knbins_dft.{png,json}
PPM z [1.5,2] · +DFT · min20 (sensitivity)	~24%	~11%	YES	..._knbins_min20_ppm_z1p5_2_dft.*

Band (primary): 392 PS / 119 drafts. Many **thin tail cells** (n < 30) — expected; read **direction**, not last bin alone.

Caveat: Old 16-quantile T_j panel showed **monotone rise** — wrong binning, not falsification of Alex’s story.

Open: Formal K/N → minimum bin-count memo (COMPASS); MLE knee fit (Act III / later).

Priority 3 — Within-team rank × T_j quartile — SHIPPED

Draft rate vs **roster percentile**, faceted by **T_j quartile**. Files: CCT_draft_rate_roster_pct_by_tj_quartile.{png,json}.

Result (2026-08-21): Q4 (highest T_j) — top roster rank **9.5%** vs mid rank **3.0%** (top > mid: **YES**). All quartiles show top rank > mid rank. Does **not** fix Å — complements P1; supports “stand out on **your** team,” especially on contenders.

Priority 4 — Interval overlap + draft (HAND17 extension)

Color drafted vs not on interval slide logic; top T_j quartile; drafted players on **leading edge** of team interval?

Priority 5 — Model-aligned diagnostics

- L_j^C vs T_j by ventile
- MLE residuals: observed – predicted draft rate by poolq_loo at **fixed A**

Act IIb — Sensitivity — MOSTLY DONE

Variant	Status	Result
+DFT subsample (PPM P1)	✓	CCT YES , fragile (~2 pp)
OBPM full + +DFT	✓	CCT NO both; appendix only
T_j secondary panel	✓	Rate rises with team mean
mg10 min10 P1 grid (18 cells)	✓	CCT_p1_grid_manifest_min10.json
PPM triptych (poolq_loo deck)	✓	CCT_draft_rate_ai_band_poolq_loo_min10_ppm_triptych.png (+ _b8)
P2b Alex board (T_j tail bins)	✓	CCT_draft_rate_fixedAi_Tj_knbins_dft.*
Alternate Å bands	✓	[2,3] full/+DFT CCT YES on poolq_loo; see min10 grid

4 — Engineering plan

New script (SCOUT)

Name (locked): sports/scripts/pass_a_congestion_conditional.py

Reuse: pass_a_empirical_bundle.py panel path, pd20_22_campaign_window, hero_gallery_paths, BDP layout helpers from bdp_ai_tj_distributions.py where sensible.

CLI sketch

```
python sports/scripts/pass_a_congestion_conditional.py \  
  --season-min 2011 --season-max 2021 \  
  --min-minutes 20 \  
  --min-team-season-games 10 \  
  --winsor-lo 0.01 --winsor-hi 0.99 \  
  --ai-lo 1.5 --ai-hi 2.0 \  
  --plot matched_pond \  
  --out-dir 3-Master_Plan/re_entry/HEROs_and_PASSes/basic_data_plots
```

[SCOUT] mg10 = min_team_season_games 10 in code (drop team-seasons with ≤10 games; keep 11+). Do not pass 11 — that is not the repo knob name.

Incremental writes: JSONL or per-plot JSON with n, drafts, rate, CI per cell — append-friendly if we add bands later (../../../../cursor/rules/incremental-writes.mdc).

Do not clobber: Locked Pass A hero PNGs in pass_a/.

HAND deck (Charles)

New deck or new section in next HAND — working title CHAR_CCT_HAND or section in PD23 deck. **Change Picture** from basic_data_plots/ and slides/Basics_data_plots_HAND/ exports. Agents **never** overwrite .pptx.

5 — Checklist (Charles checkboxes only)

How to use: Charles changes [] → [x] and fills Proof. SCOUT/COMPASS update Status rows in §6.

Phase 0 — Green light

Done	Item	Proof
[]	Charles read porch doc + this campaign plan	Date
[x]	SCOUT reviewed this campaign plan	20260821_SCOUT_CCT_campaign_review.md
[x]	Priority 1 spec locked: Å band, axis (poolq_loo vs T_j), season window	poolq_loo · z ∈ [1.5, 2.0] · 2011–2021 · mg10 min20 · winsor 0.01–0.99
[x]	Charles says go on script build	2026-08-21 — Alex briefed; green-light Priority 1

Phase 1 — Priority 1 figure (Act II core)

Done	Item	Proof
[x]	pass_a_congestion_conditional.py exists, smoke test runs	python sports/scripts/pass_a_congestion_conditional.py --plot matched_pond ... exit 0 (2026-08-21)
[x]	Matched Å × poolq_loo PNG + JSON on disk	basic_data_plots/CCT_draft_rate_ai_band_poolq_loo.{png,json}
[x]	JSON cell counts reviewed — no empty claims from n < 10 cells	All 16 bins n≈127–128; band n=2,035 / 178 drafts; CCT=NO (Jackal 10.8% > Squid 7.3%)
[]	Charles eyeballs figures — Squid/Jackal test + sweep story	See §3 scorecard
[x]	COMPASS slide caption draft (score ≠ select language)	§7 (two-panel + P3)

Phase 2 — Act II expansion

Done	Item	Proof
[]	Priority 2 heatmap (full + +DFT if feasible)	Parked — optional
[x]	Priority 2b Alex board plot (fixed Å × T_j, K/N tail bins, +DFT)	CCT_draft_rate_fixedAi_Tj_knbins_dft.{png,json} · downturn YES
[x]	PPM triptych (poolq_loo deck)	...trptych.png , ...trptych_b8.png
[x]	Priority 3 roster percentile faceted	CCT_draft_rate_roster_pct_by_tj_quartile.{png,json}
[x]	+DFT subsample rerun of Priority 1	CCT_draft_rate_ai_band_poolq_loo_dft.*
[x]	OBPM robustness (full + +DFT)	...obpm.* , ...obpm_dft.* — appendix only

Phase 3 — Act III (Alex-ready)

Done	Item	Proof
[]	HAND deck section assembled	.pptx path
[]	Alex paragraph rehearsed (§8 below)	PD # or date
[]	BINDING rule visible on one slide or speaker note	Slide #
[]	Campaign doc Status updated to Closed or Act III complete	This file

6 — Ownership & delegation

Task	Owner	Delegate?
Campaign sequencing, checklist, Alex wording	COMPASS	—
Script build, plots, JSON spot-checks	SCOUT	—
Green-light specs, HAND assembly, checkboxes	Charles	—
Slide visual design, Change Picture	Charles	—
Theory / manuscript integration later	VECTOR	When Act III done

SCOUT first deliverable when green-lit: Priority 1 PNG + JSON + 5-line README stub in `basic_data_plots/CCT_README.md` (what panel, what band, what to look for).

COMPASS standing job: Keep this file’s **Status** and Phase tables current after each SCOUT drop; file `SCOUT_report_to_COMPASS.md` summary when a phase closes.

7 — Slide / caption language (COMPASS draft — edit in HAND)

Slide A — P1 full vs +DFT (side-by-side)

Title: *Same talent, different pond — league vs draft-ecosystem*

Caption (binding-safe):

*At fixed PPM $z \in [1.5, 2.0]$: **full panel** — thicker teammate ponds $\rightarrow$ **higher** draft rates (Jackal > Squid). **+DFT subsample** (programs with ≥ 1 pick in window) — sign **flips**: mid-pond slightly > top-pond (~23% vs ~21%), overlapping uncertainty. Not proof of environment effects; consistent with congestion **ranking** story **within** draft ecosystems only.*

Slide B — Priority 3 Q4

Title: *On contenders, draft mass sits at the top of the roster*

Caption:

Draft rate vs within-team roster percentile, faceted by team talent quartile. Q4 (highest T_j): top rank ~9.5% vs mid ~3.0%. Does not fix $\hat{A}$ — shows “big fish on **this team” matters, especially on elite rosters.**

Speaker note for Alex:

Fix- A_i was the right move. Full league confounds program quality with pond thickness. The whisper may live in draft-ecosystem programs on PPM, plus within-team rank on contenders — not the old marginal hero curve.

8 — One paragraph for Alex (read aloud — post P2b)

*We closed Act I and built the fix- A_i microscope you asked for. On the **full** NCAA panel at matched talent, players in **thicker** teammate ponds get drafted **more**, not less — elite-program confound. When we restrict to **programs that actually produce draft picks**, the sign **flips** on PPM: mid-pond slightly beats top-pond at [1.5,2], and at **[2,3]** the top-pond ventile **cliffs** — that’s the triptych. On **team talent** with your tail binning — not blind ventiles — we see the **flat-then-down** shape you drew on the board: ~34% draft rate in the mid team-talent bins, ~8% in the elite tail, same fixed player slice. OBPM goes the other way — appendix only. On **contender rosters**, draft mass concentrates at the **top of the team**. That’s our honest NCAA picture: three magnifiers — peer pond, team pond, within-team rank — not the old marginal hero dip.*

9 — Do-not-do (save campaign time)

1. **More ESPN seasons alone** — unlikely to reveal CCT (you + Alex agree).
2. **Another marginal poolq_loo ventile run** hoping bin 16 dips.
3. **Promoting OBPM/BPM to canonical hero** — appendix magnifier only.
4. ****Confusing T_j with poolq_loo**** in slides or prose.
5. **Claiming Army tiny K/N** — wrong; Army screams via macro tail drop.
6. **Claiming CCT confirmed from +DFT PPM alone** — ~2 pp, overlapping CIs, noisy ventiles.
7. **MERGE score + select + environment** in one sentence — BINDING violation.

10 — Collaboration model (three of you + two agents)

What’s been working

- Porch reading as **single narrative**; technical memos as supplements.
- SCOUT **review file** with spot-checks (not silent edits).
- Charles **checkboxes only Charles touches**.

Tighter loop for this campaign (recommended)

Artifact	Purpose	Who updates
This file	Operational source of truth — priorities, checklist, specs	COMPASS
Porch reading	Stable story for Ginger / re-entry	Frozen unless science changes
SCOUT_report_to_COMPASS.md	End-of-phase: what shipped, paths, blockers	SCOUT after each Priority
YYYYMMDD_SCOUT_CCT_campaign_review.md	One-time sign-off on this plan (like porch review)	SCOUT
Question files	Only when blocked: YYYYMMDD_COMPASS_to_SCOUT_questions.md	COMPASS

What to stop doing

- Parallel memos that re-merge the same plot menu (porch + campaign + joint memo is enough).
- SCOUT building without Charles **Phase 0 green-light** on $\hat{A}$ band and axis.
- Long chat threads as spec — **spec lives in §3–4 of this file**.

Charles’s workflow

- Send SCOUT this file.
- Lock Priority 1 spec (one message: band + axis).
- SCOUT builds → drops PNG/JSON → pings in report.
- You assemble HAND → brief Alex → mark checklist.

Optional later: Mirror a slim `.plan.md` to `plans/` if you want PDF via `convert_single_md_to_pdf.sh` — not required for day-one execution.

11 — Open decisions (Charles picks — defaults in parentheses)

Decision	Locked (2026-08-21)	Alternatives (later)
$\hat{A}$ band for Priority 1	$z \in [1.5, 2.0]$ ✓	Top decile; [1.0, 1.5] if bins thin
Primary pond axis	<code>poolq_loo</code> ventiles ✓	T_j ventiles (secondary panel)
Alex board X-axis (P2b)	**T_j with tail bins** ✓ shipped	Blind 16-ventile T_j (wrong shape — superseded)
Ventile count	16 (match hero)	10 wider bins at top
First HAND target	New CCT section	Append to next PD HAND

12 — Status log

Date	Event
2026-08-20	Act I BDP closed; porch reading published
2026-08-21	SCOUT porch review complete; Army/K/N clarifications merged
2026-08-21	Charles briefed Alex on fix $\hat{A}_i$ — Alex excited
2026-08-21	This campaign plan published — awaiting Charles green-light
2026-08-21	SCOUT campaign review complete — four engineering patches in §4
2026-08-21	Charles green-lit Priority 1 — <code>poolq_loo</code> , default $\hat{A}$ band; SCOUT building <code>pass_a_congestion_conditional.py</code>
2026-08-21	Act IIb sweep — +DFT PPM CCT YES (~2 pp); OBPM NO ; T_j rises; P3 Q4 top 9.5% vs mid 3.0%; see §3 scorecard
2026-08-21	COMPASS updated campaign plan + §13: Priority 2 heatmap next
2026-08-21	P1 min10 grid (18 cells) + PPM triptych (16- and 8-bin)
2026-08-21	P2b shipped — Alex board T_j tail bins; downturn YES (+DFT z [2,3] primary)
2026-08-21	Alex: on track; story framing + paper outline after next week

13 — COMPASS sequencing (current)

Order	Plot	Status
1	P1 matched $\hat{A} \times \text{poolq_loo}$ (full + +DFT grids min10/min20)	✓ Done
2	P3 roster percentile $\times T_j$ quartile	✓ Done
3	PPM triptych (poolq_loo deck)	✓ Done
4	P2b Alex board — fixed $\hat{A} \times **T_j^{**}$ (+DFT, tail bins)	✓ Done — PD25
5 (NOW)	Act III — HAND deck + paper outline/framing	Alex: after next week
—	P2 heatmap	Parked (optional)
—	P4 interval overlap + draft	Optional HAND enrich
—	PD24 BPM hero rerun	Appendix — when bandwidth

Do not: Claim CCT confirmed from +DFT alone; promote OBPM to canonical; revert to marginal hero bin-16 dip hunting.

Repo hygiene: COMPASS checks `CCT_README.md`, `CCT_Campaign_Plan.md`, and `basic_data_plots/CCT_*` at start of each Charles chat. SCOUT updates `CCT_README.md` on every drop.

Act II empirical core **closed** (2026-08-21 evening). HAND lead: **triptych** + **P2b** + P3 Q4. **Act III:** story + paper outline (Alex). Plain English: `_DISPOSABLE_PD25_Alex_board_for_dummies.md`.
— COMPASS · 2026-08-21 evening · Act III next